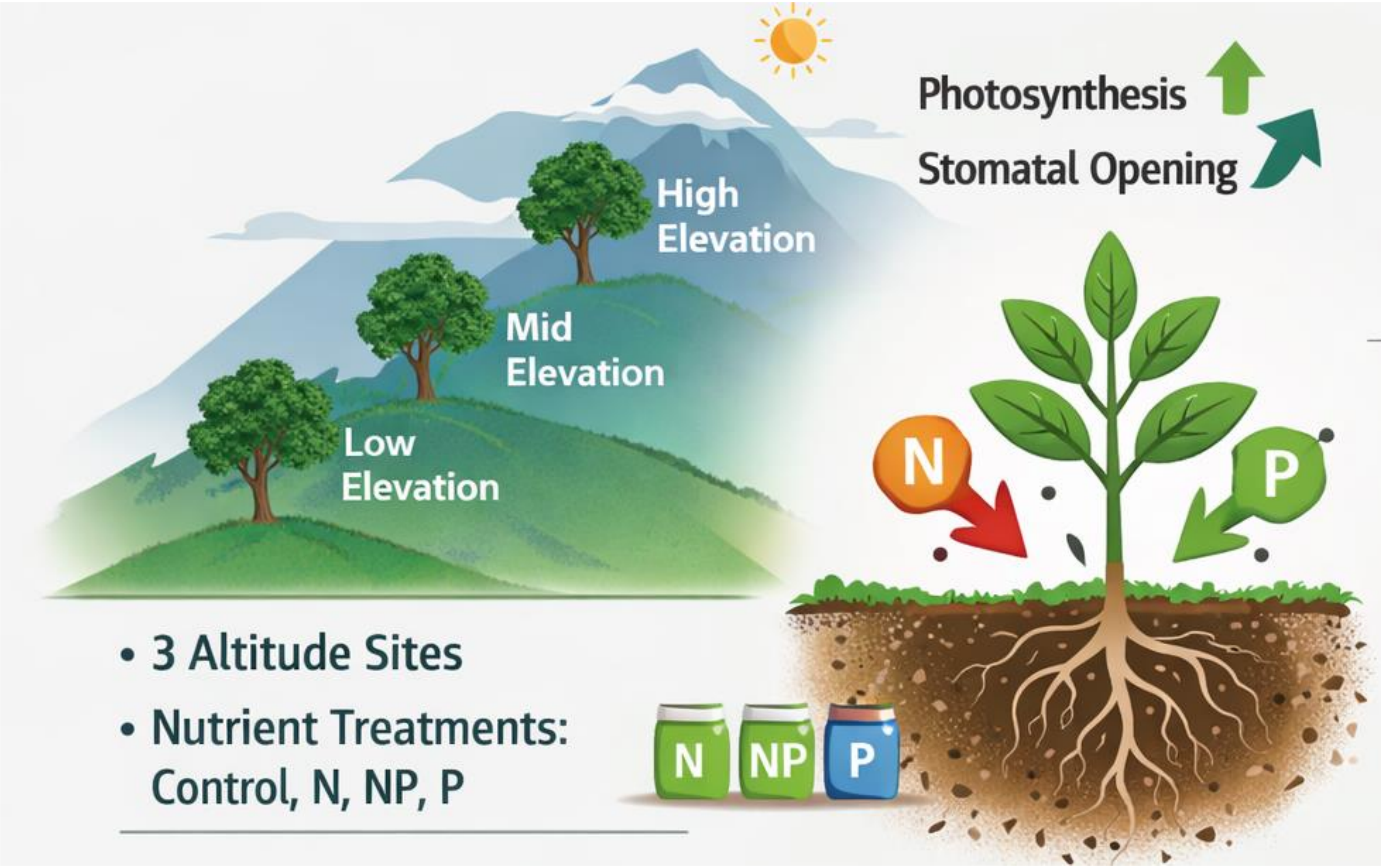
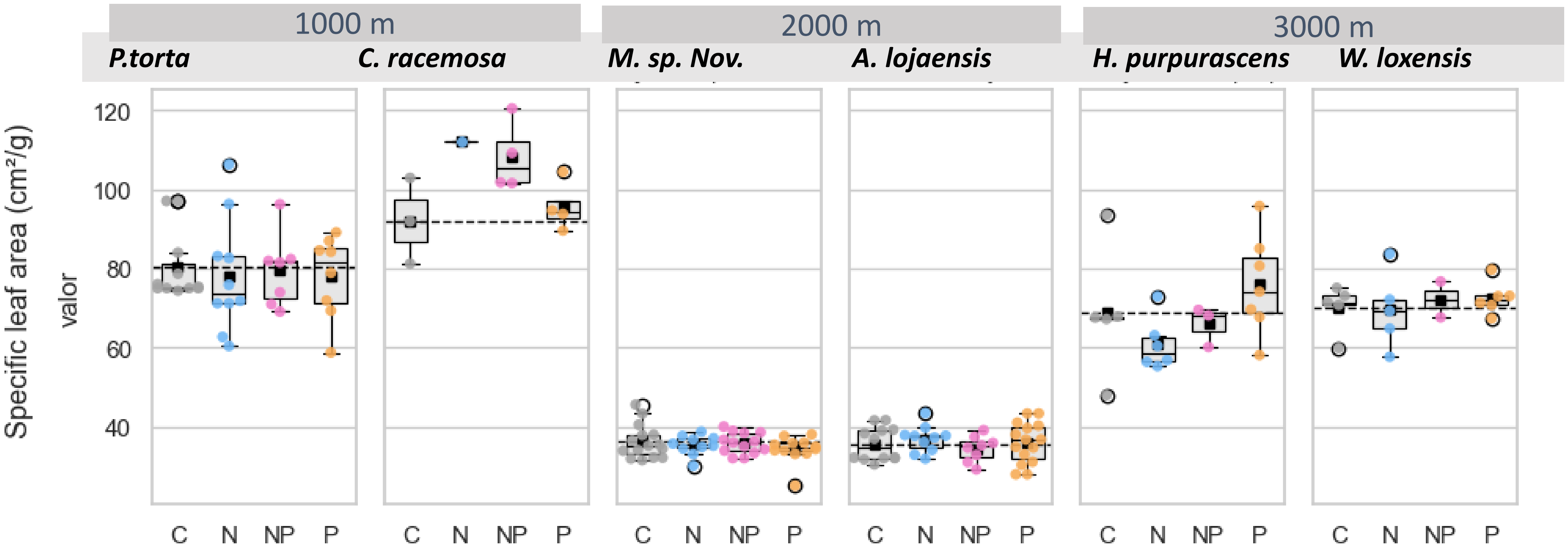

Photosynthesis
Stomatal Opening
High Elevation
Mid Elevation
Low Elevation
N
P
• 3 Altitude Sites
• Nutrient Treatments: Control, N, NP, P
N
NP
P

- *RQ1. Does moderate nutrient addition affect intrinsic water-use efficiency (iWUE) after 15 years of fertilization across treatments in an elevational gradient?*
- *RQ2. What is the effect of moderate nutrient addition to leaf nutrients (Nmass and Narea) and key leaf structural traits (SLA) after 15 years of fertilization across treatments in an elevational gradient?*
- iWUE does not show changes per treatments (slide 5) although there are differences between groups for leaf nutrients (ex. Narea, or N(mg/g)). (slide 6,7,8). SLA also does not change consistenly just *C. racemosa (*slide 9*)*
- Responses change per specie -> Working on a chart of species characteristics.
- Species at 2000 asl and 3000 asl more similar in their responses tan at 1000. Higher iWUE at mid and higher elevations associated with conservative traits (smaller SLA, higher N content).

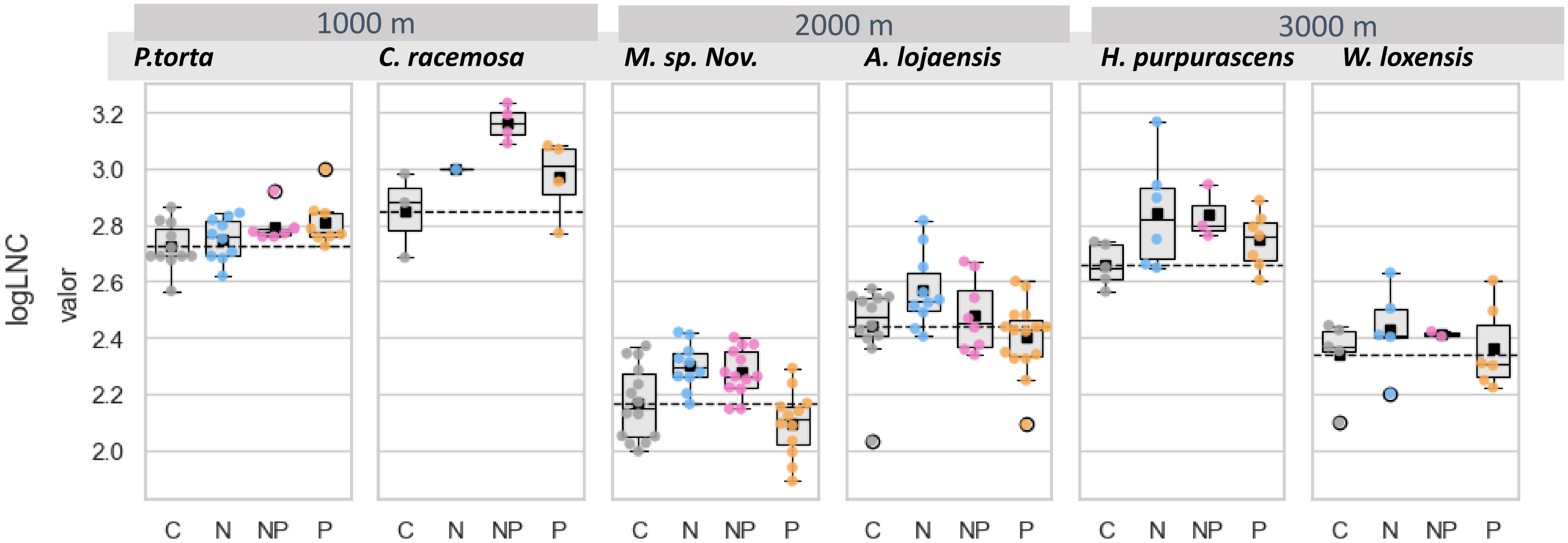
		C	N	NxP	P
elevation	especie				
1000	*Clarisia racemosa*	3	1	4	4
	Pouteria torta	11	10	7	8
2000	*Alchornea lojaensis*	12	9	8	15
	Myrcia sp nov	15	10	13	11
3000	*Hedyosmun purpurascens*	5	6	3	7
	Weinmania loxensis	5	5	2	6

			δ13C (‰ v.s.V-PDB)		iWUE (μmol/mol)		Specific leaf area (cm²/g)		LCC		LNC		LPC		N:P	
			mean	**std**	**mean**	**std**	**mean**	**std**	**mean**	**std**	**mean**	**std**	**mean**	**std**	**mean**	**std**
1000	**Clarisia racemosa**	**C**	-30.63	0.42	96.61	5.19	92.05	10.83	421.37	14.36	17.39	2.56	0.79	0.05	21.87	2.18
		N	-31.8		82.22		112.06		447.15		20.06		0.96		20.86	
		NP	-30.88	0.36	93.6	4.43	108.27	8.87	430.5	14.05	23.65	1.5	1.15	0.14	20.83	3.48
		P	-29.84	1.24	106.26	15.19	95.65	6.27	438.34	12.74	19.62	2.69	1.17	0.17	16.87	2.64
	Pouteria torta	**C**	-31.55	0.76	85.26	9.34	80.27	8.75	478	5.31	15.29	1.26	0.65	0.12	24.45	5.16
		N	-31.27	0.87	99.8	22.43	78.19	14.39	480.5	6.22	15.69	1.16	0.64	0.2	25.95	5.44
		NP	-31.74	1.35	82.99	16.58	79.48	9.22	477.95	5.16	16.38	0.98	0.73	0.07	22.58	1.81
		P	-31.27	0.93	88.76	11.48	78.02	10.48	479.21	5.51	16.68	1.52	0.77	0.13	22.47	5.11
2000	**Alchornea lojaensis**	**C**	-26.51	0.89	146.94	10.79	35.67	4.22	489.94	9.19	11.61	1.5	0.78	0.12	15.07	2.24
		N	-26.57	1.13	146.26	13.81	36.94	3.38	487.69	10.08	13.14	1.82	0.79	0.15	16.83	1.83
		NP	-26.77	1.45	143.79	17.65	34.5	3.27	490.56	10.44	12.04	1.6	1.32	0.33	9.59	2.28
		P	-27.06	1.51	140.31	18.37	36	5.12	486.93	5.09	11.1	1.37	1.24	0.28	8.94	2.79
	Myrcia sp nov	**C**	-27.79	1.15	131.33	14.07	36.24	4.23	472.37	9.44	8.82	1.17	0.43	0.17	23.28	4.17
		N	-27.24	0.92	138.06	11.23	35.42	2.48	470.86	6.79	9.99	0.83	0.43	0.03	23.54	2.22
		NP	-27.27	1	137.71	12.18	35.88	2.64	464.64	6.03	9.79	0.81	0.88	0.36	12.42	3.68
		P	-27.34	0.66	136.92	8.09	34.37	3.3	469.88	9.99	8.19	0.95	1.7	0.66	6.34	5.05
3000	**Hedyosmun purpurascens**	**C**	-26.81	1.44	143.37	17.6	68.88	16.18	438.02	6.14	14.31	1.09	0.84	0.08	17.17	1.02
		N	-26.35	2.06	148.86	25.07	60.87	6.61	434.85	10.11	17.47	3.63	0.8	0.05	21.78	4.15
		NP	-24.85	1.27	167.17	15.36	65.98	5.04	442.68	6.62	17.08	1.68	1.07	0.08	15.94	0.42
		P	-27.29	1.59	137.48	19.36	75.91	12.42	436.7	5.99	15.65	1.54	1.42	0.44	11.12	4.43
	Weinmania loxensis	**C**	-27.7	0.27	132.49	3.29	70.1	5.89	444.08	7.83	10.43	1.34	0.56	0.05	18.65	1.67
		N	-28.06	0.9	128.13	10.95	69.55	9.63	445.46	14.2	11.47	1.77	0.61	0.05	18.89	1.99
		NP	-29.03	0.79	116.22	9.7	72.21	6.39	460.1	9.12	11.17	0.14	0.72	0.07	15.57	1.25
		P	-28.38	1.13	124.17	13.8	72.58	4.07	440.55	11.17	10.73	1.7	1.64	0.44	7.31	3.43

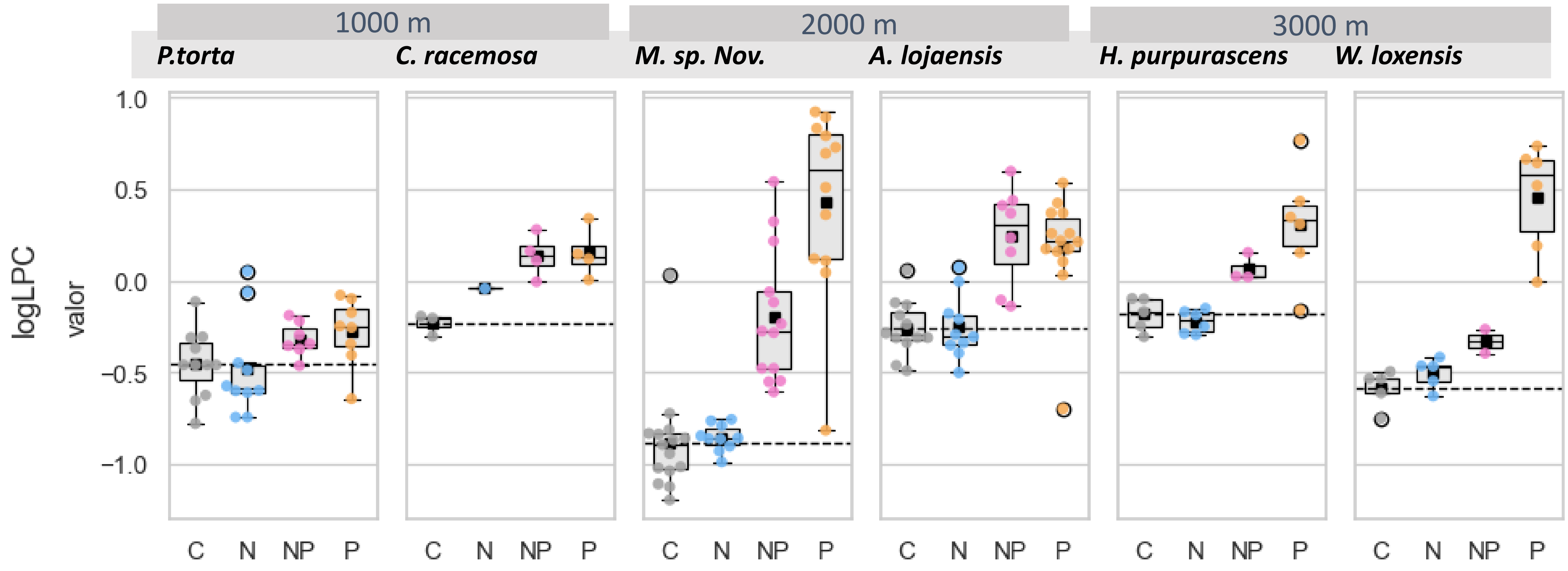
Main figures



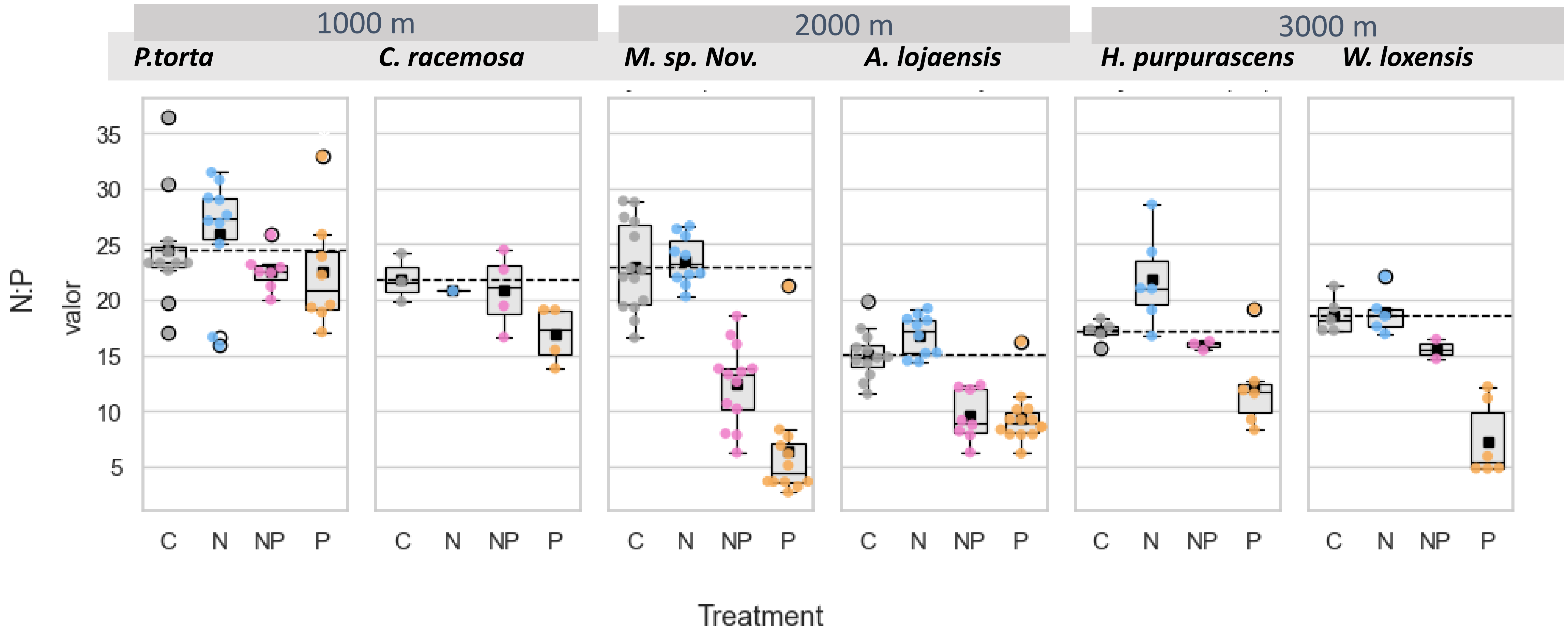
term	(Intercept)	(Intercept)	SUBN	SUBN	SUBNP	SUBNP	SUBP	SUBP	SD (Intercept Bloque)	SD (Observations)	Num.Obs.	R2 Marg.	R2 Cond.	AIC
statistic	estimate	std.error	estimate	std.error	estimate	std.error	estimate	std.error	estimate	estimate				
SLA | 1000 | Pouteria	80.265***	(3.333)	-2.075	(4.829)	-0.783	(5.344)	-2.247	(5.136)	0.000	11.053	36	0.008		265.3
SLA | 1000 | Clarisia	92.055***	(5.532)	24.498*	(7.724)	14.369*	(4.867)	7.687	(5.135)	7.217	6.303	12	0.374	0.729	71.2
SLA | 2000 | Alchornea	35.554***	(1.365)	1.260	(1.762)	-1.368	(1.875)	0.383	(1.592)	1.357	4.087	45	0.038	0.133	255.7
SLA | 2000 | Myrcia	36.240***	(0.859)	-0.816	(1.359)	-0.361	(1.261)	-1.871	(1.289)	0.000	3.329	50	0.045		263.2
SLA | 3000 | Hedyosmum	68.883***	(5.136)	-8.013	(6.954)	-2.907	(8.387)	7.024	(6.725)	0.000	11.484	21	0.222		149.7
SLA | 3000 | Weinmannia	70.102***	(3.009)	-0.552	(4.255)	2.106	(5.629)	2.477	(4.074)	0.000	6.728	18	0.040		110.8



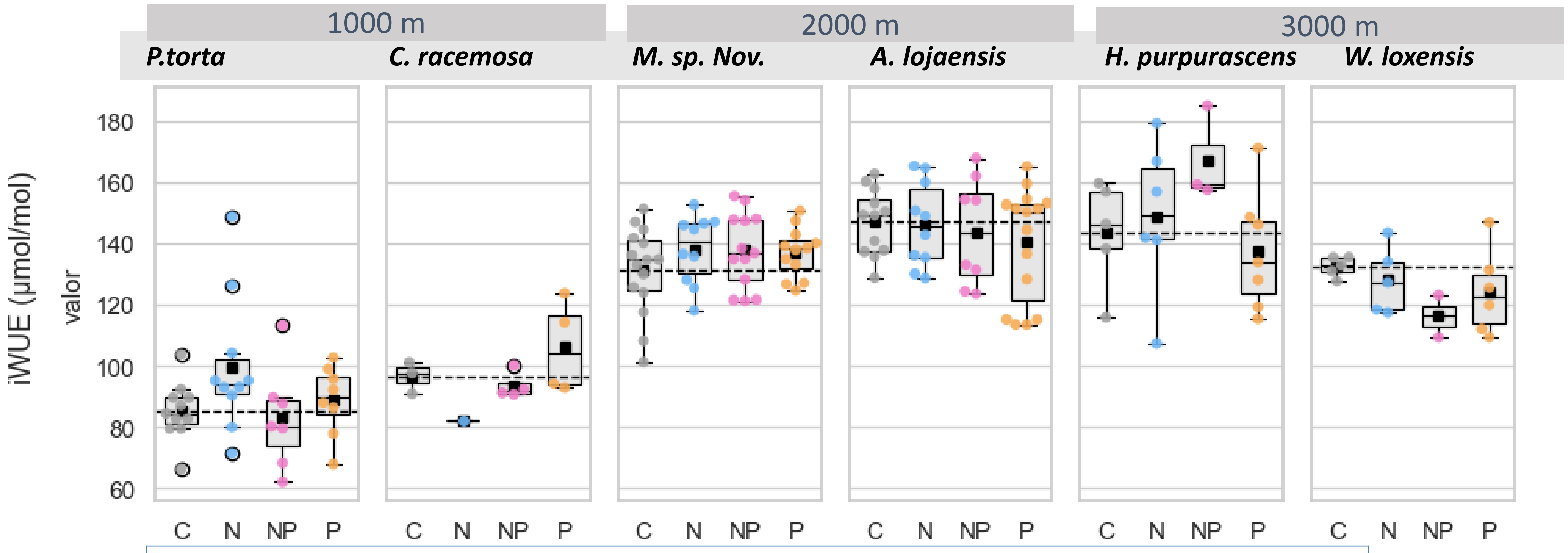
	(Intercept)	(Intercept)	N	N	NP	NP	P	P	SD (Intercept Bloque)	SD (Observations)	Num.Obs.	R2 Marg.	R2 Cond.	AIC
Modelo	estimates	estimates.1	estimates.2	estimates.3	estimates.4	estimates.5	estimates.6	estimates.7	estimates.8	estimates.9	gof	gof.1	gof.2	gof.3
logNmass | 1000 | Pouteria	2.728***	(0.030)	0.021	(0.036)	0.049	(0.036)	0.082*	(0.034)	0.038	0.071	36	0.132	0.324	-54.8
logNmass | 1000 | Clarisia	2.849***	(0.070)	0.150	(0.141)	0.313*	(0.093)	0.120	(0.093)	0.000	0.122	12	0.520		4.9
logNmass | 2000 | Alchornea	2.442***	(0.039)	0.125*	(0.058)	0.038	(0.061)	-0.043	(0.052)	0.000	0.134	45	0.182		-26.6
logNmass | 2000 | Myrcia	2.168***	(0.032)	0.132**	(0.043)	0.102*	(0.041)	-0.071+	(0.041)	0.031	0.104	49	0.350	0.405	-52.1
logNmass | 3000 | Hedyosmum	2.648***	(0.065)	0.201**	(0.065)	0.163+	(0.078)	0.088	(0.062)	0.089	0.106	21	0.243	0.557	-5.4
logNmass | 3000 | Weinmannia	2.338***	(0.064)	0.092	(0.091)	0.076	(0.121)	0.026	(0.087)	0.000	0.144	18	0.067		3.2



	(Intercept)	(Intercept)	N	N	NP	NP	P	P	SD (Intercept Bloque)	SD (Observations)	Num.Obs.	R2 Marg.	R2 Cond.	AIC
logLPC | 1000 | Pouteria	-0.446***	(0.076)	-0.038	(0.093)	0.121	(0.096)	0.178+	(0.091)	0.086	0.189	36	0.153	0.298	7.5
logLPC | 1000 | Clarisia	-0.233*	(0.067)	0.194	(0.134)	0.370**	(0.088)	0.386**	(0.088)	0.000	0.116	12	0.684		4.1
logLPC | 2000 | Alchornea	-0.257**	(0.081)	-0.019	(0.092)	0.489***	(0.098)	0.428***	(0.085)	0.106	0.213	44	0.491	0.592	14.9
logLPC | 2000 | Myrcia	-0.883***	(0.090)	0.027	(0.143)	0.687***	(0.132)	1.315***	(0.135)	0.000	0.349	50	0.707		55.8
logLPC | 3000 | Hedyosmum	-0.183*	(0.082)	-0.039	(0.111)	0.249+	(0.134)	0.492***	(0.111)	0.000	0.183	20	0.622		9.4
logLPC | 3000 | Weinmannia	-0.585***	(0.086)	0.081	(0.122)	0.255	(0.162)	1.043***	(0.117)	0.000	0.193	18	0.858		11.4

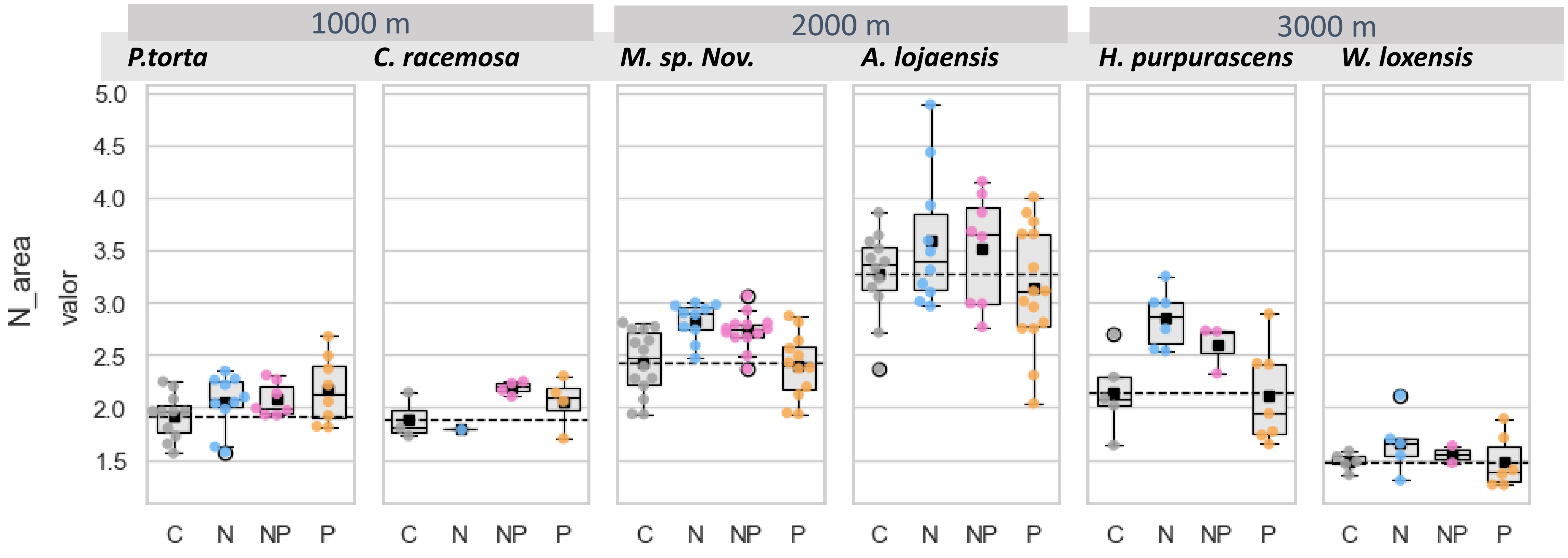


	(Intercept)	N	NP	P	Num.Obs.	R2 Marg.	R2 Cond.	AIC
Leaf N:P | 1000 | Pouteria	24.474***	1.444	-1.987	-2.163	36	0.087	0.140	211.6
Leaf N:P | 1000 | Clarisia	21.867***	-0.995	-1.076	-4.953+	12	0.356	0.374	55.5
Leaf N:P | 2000 | Alchornea	15.015***	2.175*	-5.308***	-5.939***	45	0.653	0.769	203.1
Leaf N:P | 2000 | Myrcia	23.224***	0.308	-11.021***	-16.982***	50	0.772	0.788	279.1
Leaf N:P | 3000 | Hedyosmum	17.046***	4.648*	-1.420	-6.056**	21	0.599	0.645	109.1
Leaf N:P | 3000 | Weinmannia	18.652***	0.234	-3.078	-11.345***	18	0.823		83.1

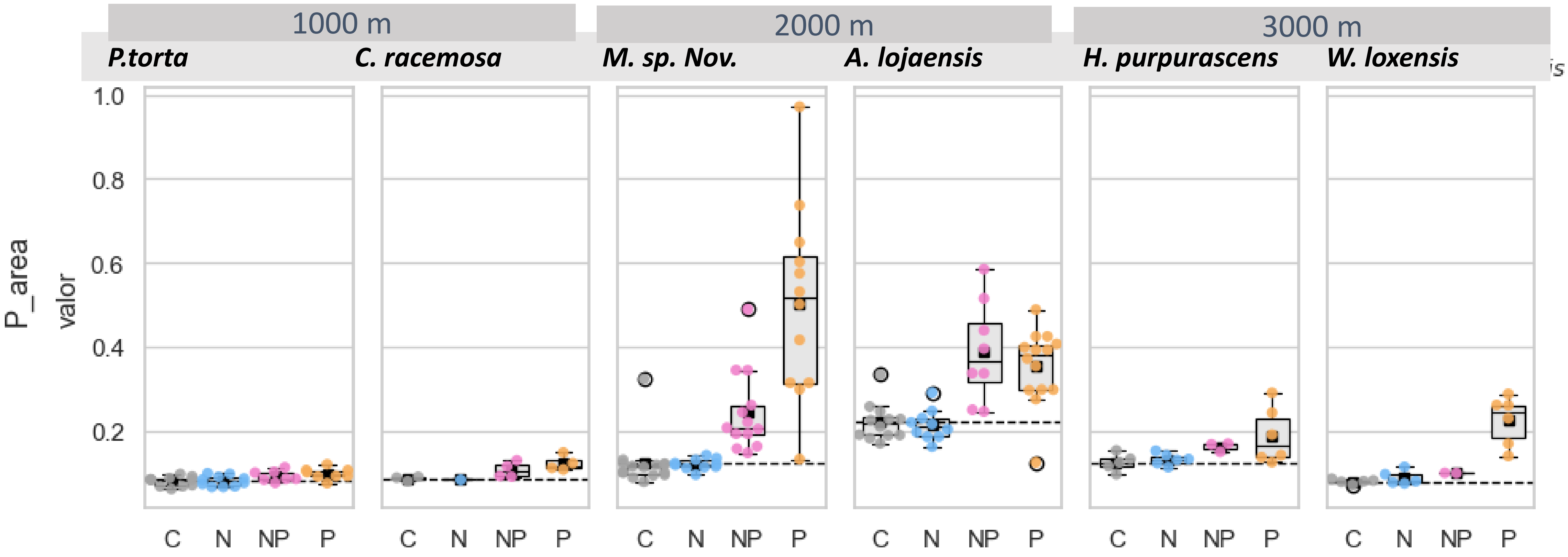


term	(Intercept)	SUBN	SUBNP	SUBP	SD (Intercept Bloque)	SD (Observations)	Num.Obs.	R2 Marg.	R2 Cond.	AIC
statistic	estimate	estimate	estimate	estimate	estimate	estimate				
iWUE | 1000 | Pouteria	85.255***	14.543*	-2.266	3.506	0.000	15.786	36	0.152		288.1
iWUE | 1000 | Clarisia	96.605***	-14.388	-3.010	9.652	0.000	10.033	12	0.353		75.5
iWUE | 2000 | Alchornea	146.959***	0.066	-2.984	-7.104	7.294	14.158	45	0.039	0.241	359.2
iWUE | 2000 | Myrcia	131.326***	6.731	6.381	5.593	0.000	11.800	50	0.057		379.7
iWUE | 3000 | Hedyosmum	143.373***	5.491	23.798	-5.890	0.000	20.438	21	0.188		169.3
iWUE | 3000 | Weinmannia	132.195***	-4.230	-14.820	-7.505	3.322	10.186	18	0.155	0.236	123.3

Narea and Parea (optional)



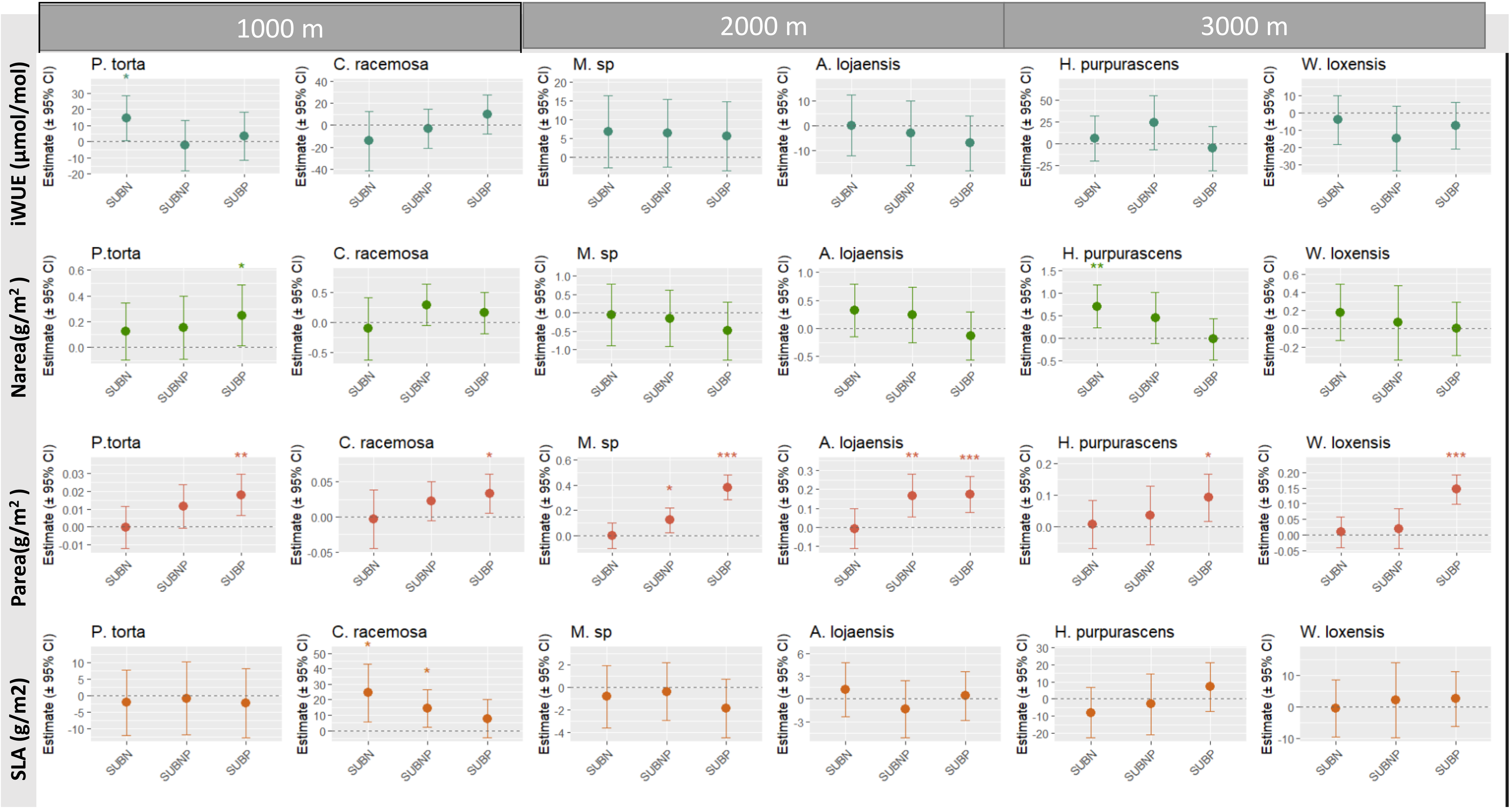
	(Intercept)	(Intercept)	N	N	NP	NP	P	P	SD (Intercept Bloque)	SD (Observations)	Num.Obs.	R2 Marg.	R2 Cond.	AIC
Modelo	estimates	estimates.1	estimates.2	estimates.3	estimates.4	estimates.5	estimates.6	estimates.7	estimates.8	estimates.9	gof	gof.1	gof.2	gof.3
Narea | 1000 | Pouteria	1.920***	(0.075)	0.127	(0.108)	0.154	(0.120)	0.250*	(0.115)	0.000	0.248	36	0.123		22.3
Narea | 1000 | Clarisia	1.892***	(0.112)	-0.102	(0.224)	0.295+	(0.148)	0.159	(0.148)	0.000	0.194	12	0.342		12.3
Narea | 2000 | Alchornea	3.272***	(0.161)	0.320	(0.231)	0.243	(0.247)	-0.133	(0.209)	0.079	0.539	45	0.106	0.125	87.9
Narea | 2000 | Myrcia	2.882***	(0.260)	-0.057	(0.410)	-0.152	(0.380)	-0.486	(0.389)	0.046	1.003	50	0.035	0.037	153.0
Narea | 3000 | Hedyosmum	2.136***	(0.170)	0.702**	(0.222)	0.448	(0.268)	-0.022	(0.214)	0.094	0.365	21	0.433	0.468	33.2
Narea | 3000 | Weinmannia	1.484***	(0.102)	0.182	(0.144)	0.069	(0.191)	-0.000	(0.138)	0.000	0.228	18	0.114		16.1



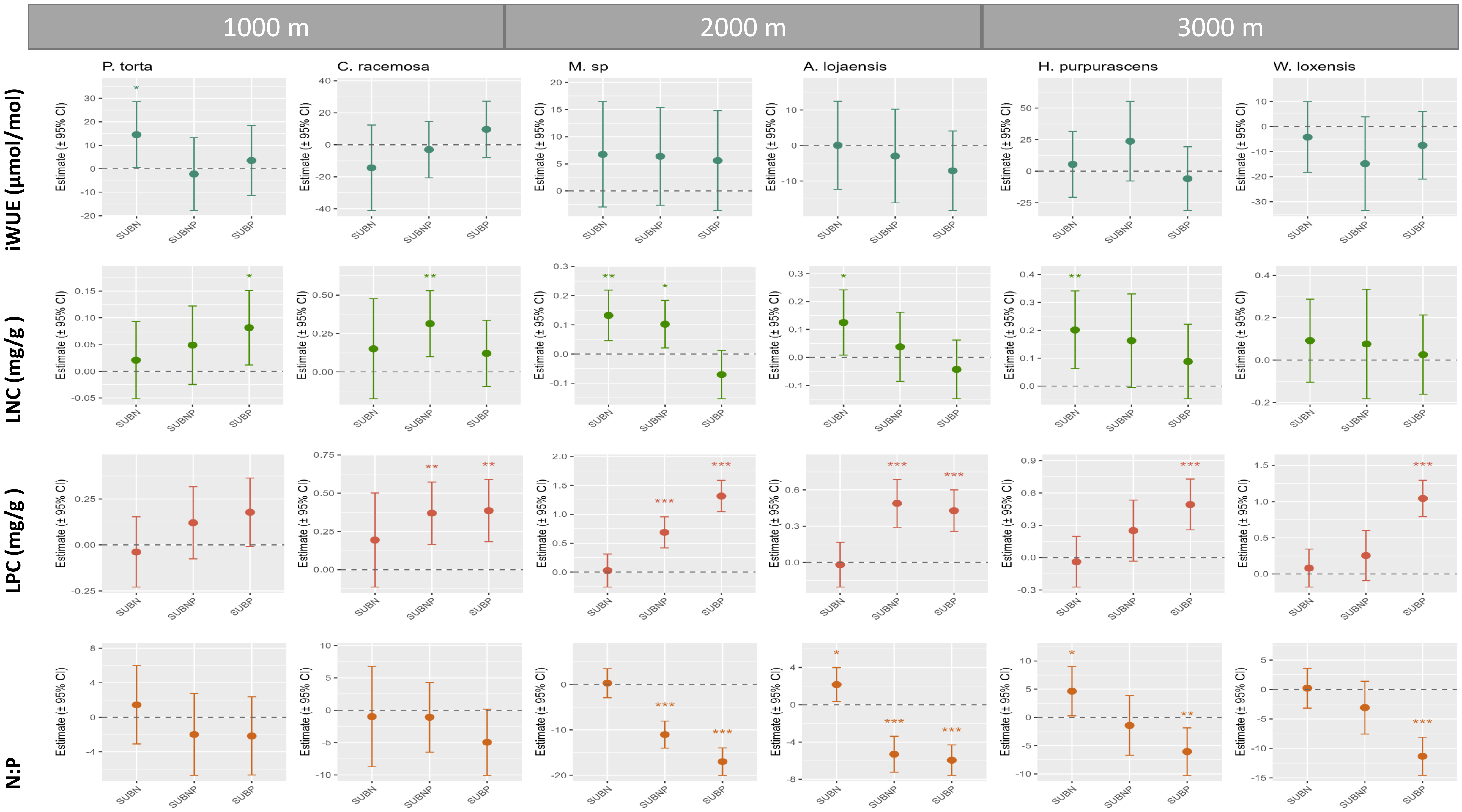
	(Intercept)	(Intercept)	N	N	NP	NP	P	P	SD (Intercept Bloque)	SD (Observati ons)	Num.Obs.	R2 Marg.	R2 Cond.	AIC
Modelo	estimates	estimates.1	estimates.2	estimates.3	estimates.4	estimates.5	estimates.6	estimates.7	estimates.8	estimates.9	gof	gof.1	gof.2	gof.3
Parea | 1000 | Pouteria	0.081***	(0.005)	-0.000	(0.006)	0.011+	(0.006)	0.018**	(0.006)	0.005	0.012	36	0.279	0.380	-170.3
Parea | 1000 | Clarisia	0.087***	(0.010)	-0.003	(0.018)	0.022+	(0.011)	0.033*	(0.012)	0.008	0.015	12	0.437	0.579	-27.6
Parea | 2000 | Alchornea	0.221***	(0.036)	-0.008	(0.052)	0.167**	(0.056)	0.174***	(0.047)	0.011	0.122	45	0.345	0.351	-34.5
Parea | 2000 | Myrcia	0.122***	(0.033)	-0.001	(0.051)	0.122*	(0.048)	0.382***	(0.049)	0.000	0.126	50	0.604		-38.1
Parea | 3000 | Hedyosmum	0.125***	(0.027)	0.008	(0.036)	0.038	(0.044)	0.093*	(0.035)	0.000	0.060	21	0.319		-29.1
Parea | 3000 | Weinmannia	0.080***	(0.016)	0.009	(0.023)	0.020	(0.030)	0.145***	(0.022)	0.000	0.036	18	0.778		-35.7

Coeficients plot

- Still figuring out if work with LNC or Narea



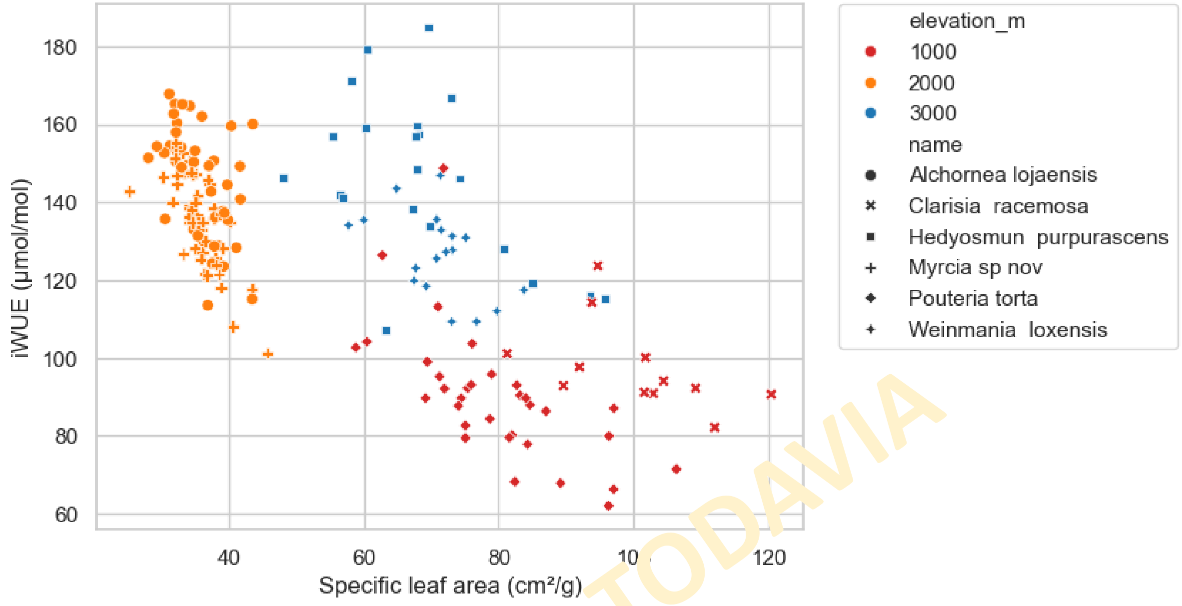
1000 m
2000 m
3000 m
iWUE (µmol/mol)
Narea(g/m²)
Parea(g/m²)
SLA (g/m2)
P. torta
C. racemosa
M. sp
A. lojaensis
H. purpurascens
W. loxensis
Estimate (± 95% CI)
SUBN
SUBNP
SUBP

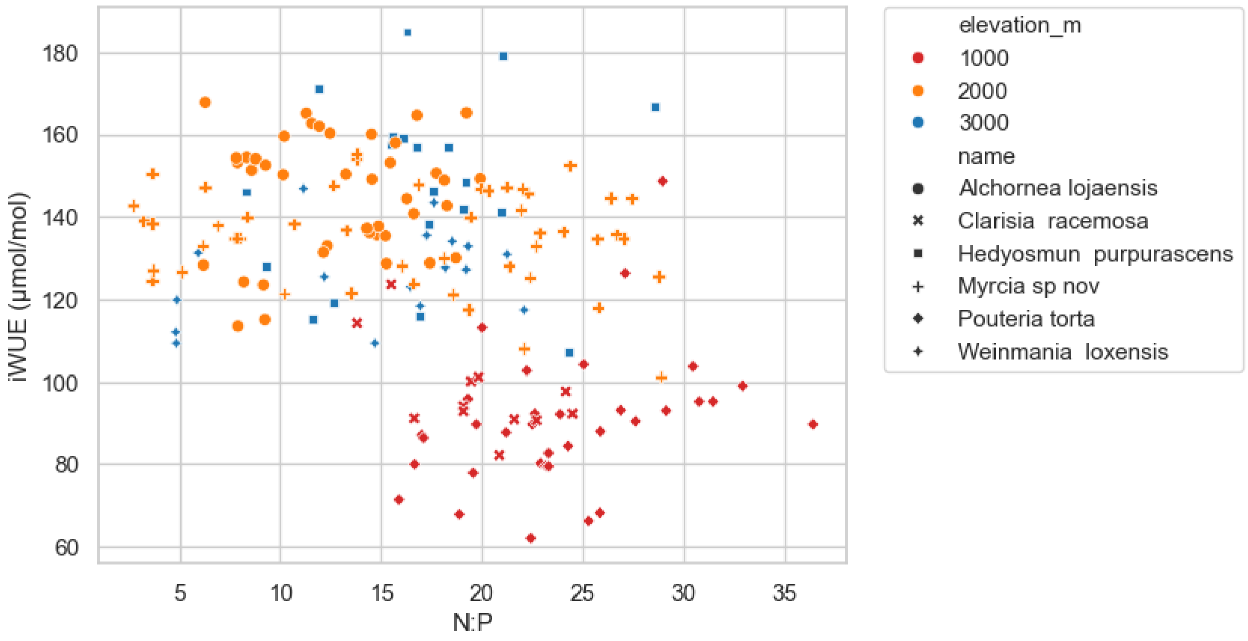


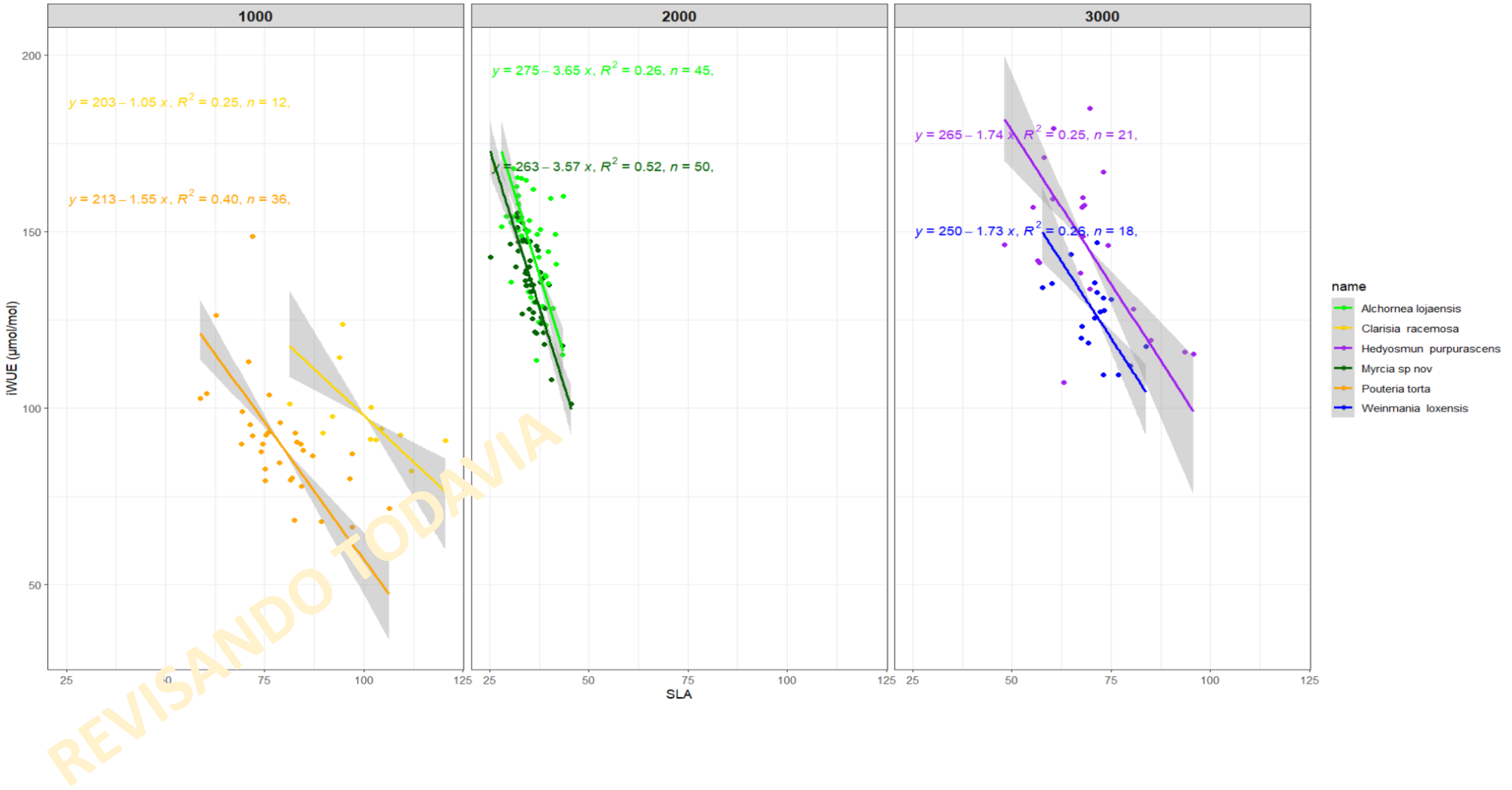
1000 m
2000 m
3000 m
P. torta
C. racemosa
M. sp
A. lojaensis
H. purpurascens
W. loxensis
iWUE (µmol/mol)
LNC (mg/g)
LPC (mg/g)
N:P
Estimate (± 95% CI)
SUBN
SUBNP
SUBP

RQ3:*What is the relationship between leaf nutrients (Nmass and Narea) and morphology (SLA) to iWUE at the different altitudes? – No treatments?*
Will explain the trends at different elevations.

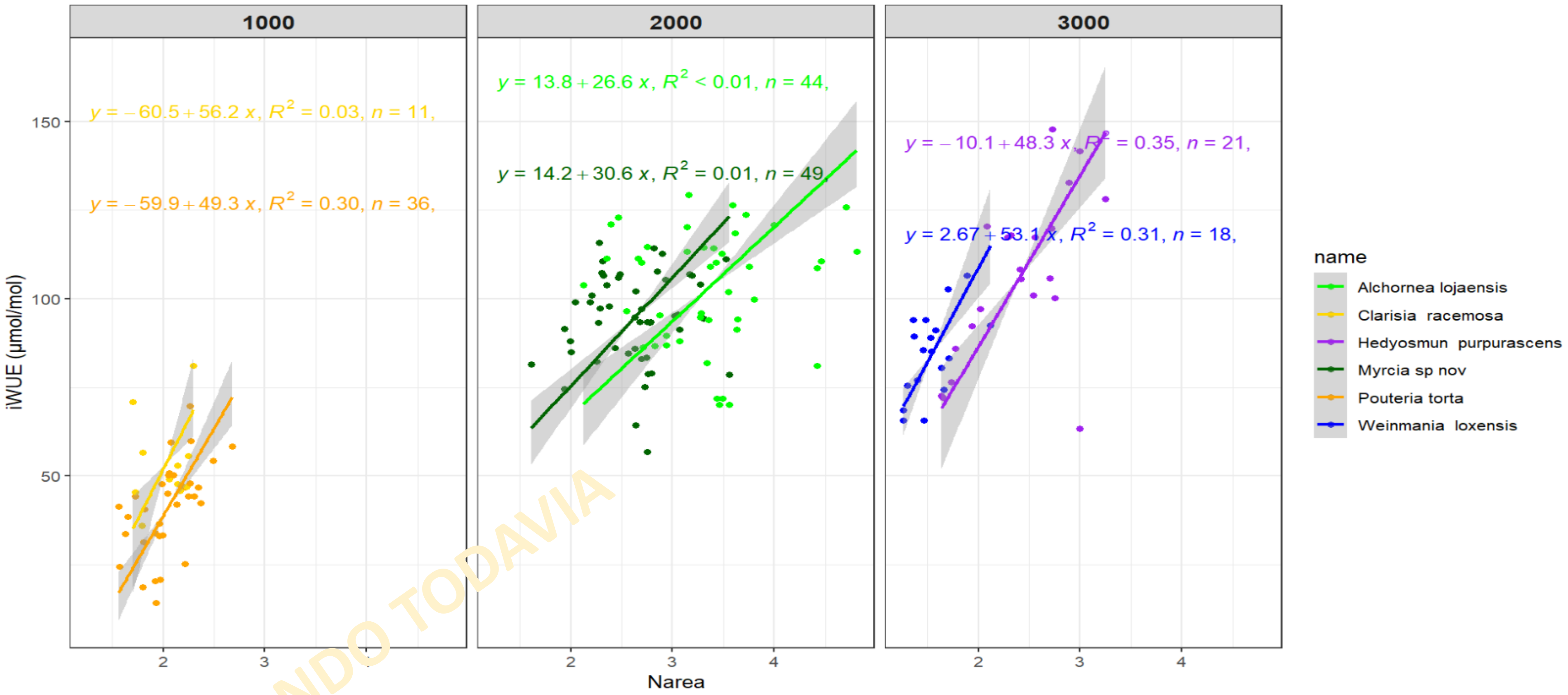
*Which analysis? SMA? Or lmer (R ~ Narea * Treat) per specie*







1000
2000
3000
y = 203 – 1.05 x, R2 = 0.25, n = 12,
y = 213 – 1.55 x, R2 = 0.40, n = 36,
y = 275 – 3.65 x, R2 = 0.26, n = 45,
y = 263 – 3.57 x, R2 = 0.52, n = 50,
y = 265 – 1.74 x, R2 = 0.25, n = 21,
y = 250 – 1.73 x, R2 = 0.26, n = 18,
iWUE (µmol/mol)
SLA
name
Alchornea lojaensis
Clarisia racemosa
Hedyosmun purpurascens
Myrcia sp nov
Pouteria torta
Weinmania loxensis
REVISANDO TODAVIA



1000
2000
3000
$y = -60.5 + 56.2\,x,\ R^2 = 0.03,\ n = 11,$
$y = -59.9 + 49.3\,x,\ R^2 = 0.30,\ n = 36,$
$y = 13.8 + 26.6\,x,\ R^2 < 0.01,\ n = 44,$
$y = 14.2 + 30.6\,x,\ R^2 = 0.01,\ n = 49,$
$y = -10.1 + 48.3\,x,\ R^2 = 0.35,\ n = 21,$
$y = 2.67 + 53.1\,x,\ R^2 = 0.31,\ n = 18,$
iWUE (µmol/mol)
150
100
50
2
3
4
Narea
name
Alchornea lojaensis
Clarisia racemosa
Hedyosmun purpurascens
Myrcia sp nov
Pouteria torta
Weinmania loxensis
REVISANDO TODAVIA

y = 38.9 + 55.8 x, R2 = 0.19, n = 12,
y = 167 – 113 x, R2 = 0.17, n = 36,
y = 192 – 46.4 x, R2 = 0.01, n = 44,
y = 120 + 18.6 x, R2 < 0.01, n = 50,
y = 210 – 60.8 x, R2 < 0.01, n = 20,
y = 145 – 19.5 x, R2 = 0.15, n = 18,
iWUE (µmol/mol)
150
100
50
0.5
1.0
1.5
2.0
2.5
LPC
name
Alchornea lojaensis
Clarisia racemosa
Hedyosmun purpurascens
Myrcia sp nov
Pouteria torta
Weinmania loxensis
STILL THINKING ABOUT THIS ONE